ME350 FA 2025 Section 002

Lab #9

Monday 12:30 PM - 2:30 PM

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Office Hours: 3:00 PM - 5:00 PM Monday (Zoom)

Agenda

- Reminders
- Mechatronic Systems Block Diagram
- Motor Friction Compensation

Reminders

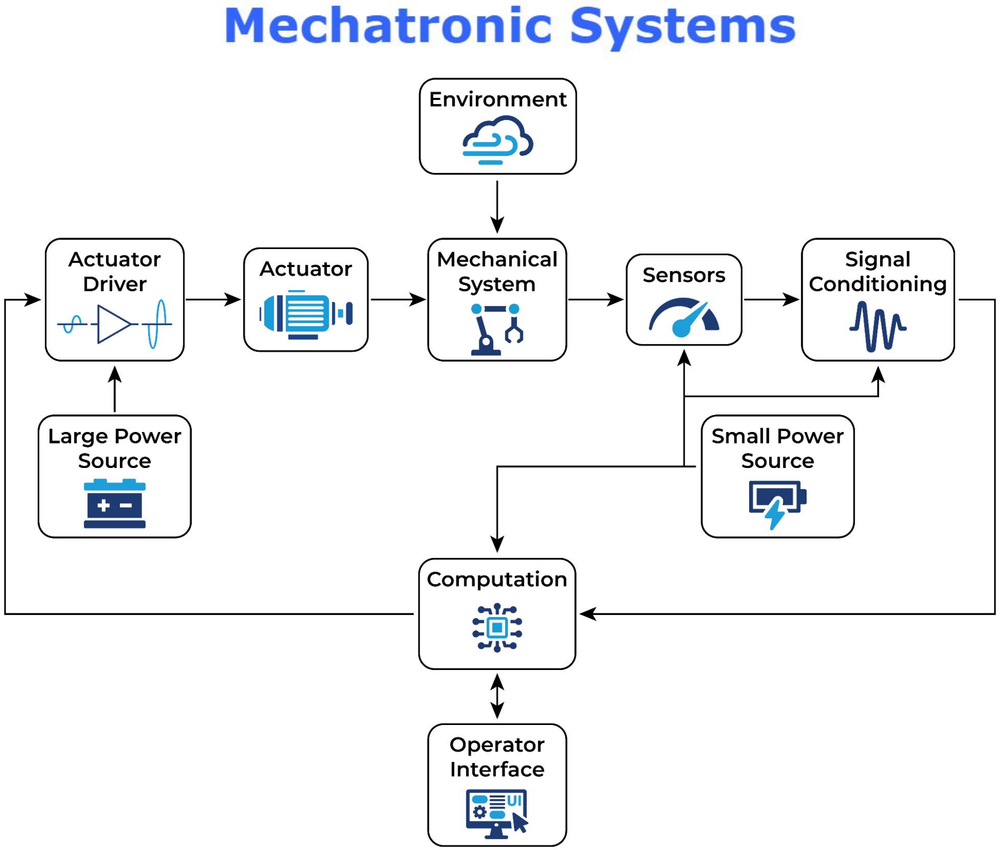
- Your transmission components should be mostly integrated by this point.
- Homework 6 is due Thursday, November 6th.
- DR2 is due Wednesday, November 5th (MONDAY LAB)

		Oct 19	Oct 20	Oct 21	Oct 22	Oct 23	Oct 24	Oct 25
9	Lecture			Lecture 15 - Sensors Part 1 (JK)		Lecture 16 - Sensors Part 2 (JK)		
	Homework					- Homework 6 Released	- Homework 5 Due (Transmissions 2)	
	Lab		Lab 8 - Sensors and Microcontrollers 1: Wiring, switches, and IR Prox sensor Lab Assignment #7 Due - Manufactured GR2 Parts		Mid semester Peer Evaluation Due	Lab 9 - Sensors and Microcontrollers 2: Motors and Encoders Lab Assignment #8 Due - Lab 8 Worksheet with screenshots	Lab 9 - Sensors and Microcontrollers 2: Motors and Encoders Lab Assignment #8 Due - Lab 8 Worksheet with screenshots	
	Project Timeline							
November		Oct 26	Oct 27	Oct 28	Oct 29	Oct 30	Oct 31	Nov 1
10	Lecture			Lecture 17 - Signals Part 1 (JK)		Lecture 18 - Signals Part 2 (JK)		
	Homework							
	Lab		Lab 9 - Sensors and Microcontrollers 2: Motors and Encoders Lab Assignment #8 Due - Lab 8 Worksheet with screenshots			Lab 10 - Sensors and Microcontrollers 3: Control Tuning Lab Assignment #9 Due	Lab 10 - Sensors and Microcontrollers 3: Control Tuning Lab Assignment #9 Due	
	Project Timeline							
		Nov 2	Nov 3	Nov 4	Nov 5	Nov 6	Nov 7	Nov 8
11	Lecture			Lecture 19 - Controls Part 1 (JK)		Lecture 20 - Controls Part 2 (JK)		
	Homework					- Homework 6 Due (Sensors) - Homework 7 Released		
	Lab		Lab 10 - Sensors and Microcontrollers 3: Control Tuning Lab Assignment #9 Due			Lab 11 - Sensors and Microcontrollers 4: State Machines Lab Assignment #10 Due	Lab 11 - Sensors and Microcontrollers 4: State Machines Lab Assignment #10 Due	
	Project Timeline		DR 2 DUE TODAY BY MIDNIGHT (Thursday Section)	DR 2 DUE TODAY BY MIDNIGHT (Friday Section)	DR 2 DUE TODAY BY MIDNIGHT (Monday Section)			

Mechatronic System Block Diagram

For any given Mechatronic System Design, should **always** start with a block diagram.

- Requires models of hardware (beyond the scope of ME 350)
  - DC motor model
  - Linkage dynamics
  - Sensor dynamics
- Understand flow of information vs flow of power.
- Understand which elements exist in the physical world and which exist in the computational world.



Open-Loop System

Open-loop systems do not use feedback.

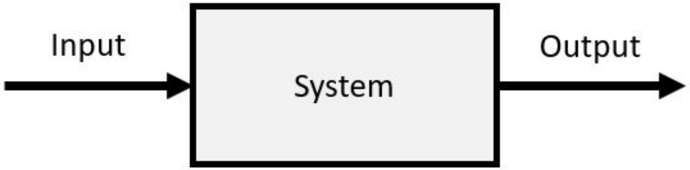
- The output signal has no effect on the input signal.

Consider a toaster.

- Provides heat for a set amount of time.
  - This is usually appropriate and sufficient for a toaster
  - What if the toast is burning? Without feedback, the toaster's input (heat/duration) does not change

In ME 350, the motor and linkage are treated as the open-loop system

- Provide sufficient voltage and the linkage will aimlessly crash into a hardstop.

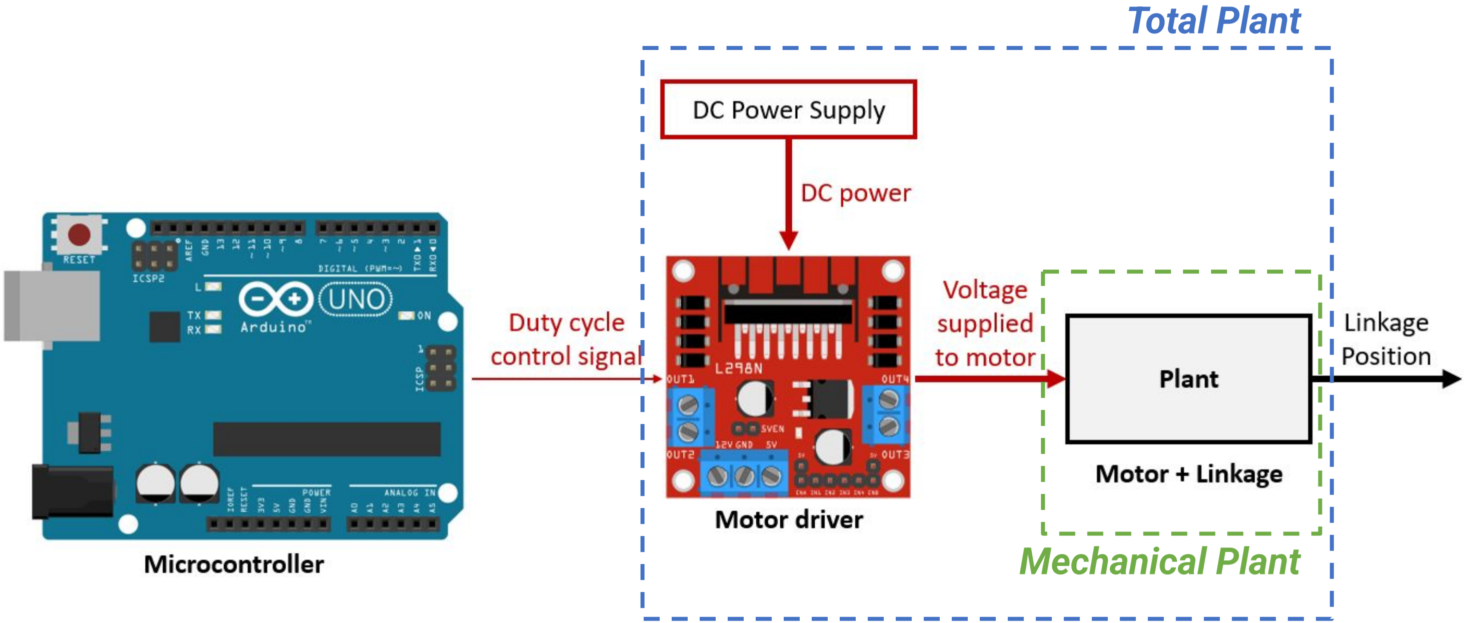


Open-Loop System

The “plant” is the motor and linkage.

- Plant input is the control command from Arduino. The output of interest is linkage position.

How do you control the position? **Closed-loop Control (Lab 10)**



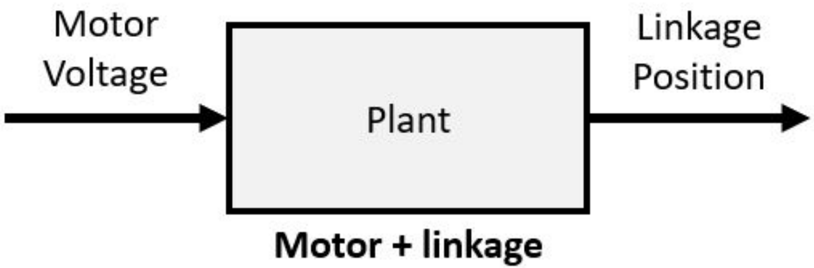
Feedforward Control - Friction Compensation

Friction always opposes motion.

- Where do the friction torques come from in this project?

How do we account for friction?

- Feedforward Control
  - If you have a **known** and **repeatable** disturbance (e.g. weight due to gravity), you can add to your control command the motor voltage that "counteracts" the disturbance.
  - **Experimentally** determine the motor voltage needed to overcome friction.



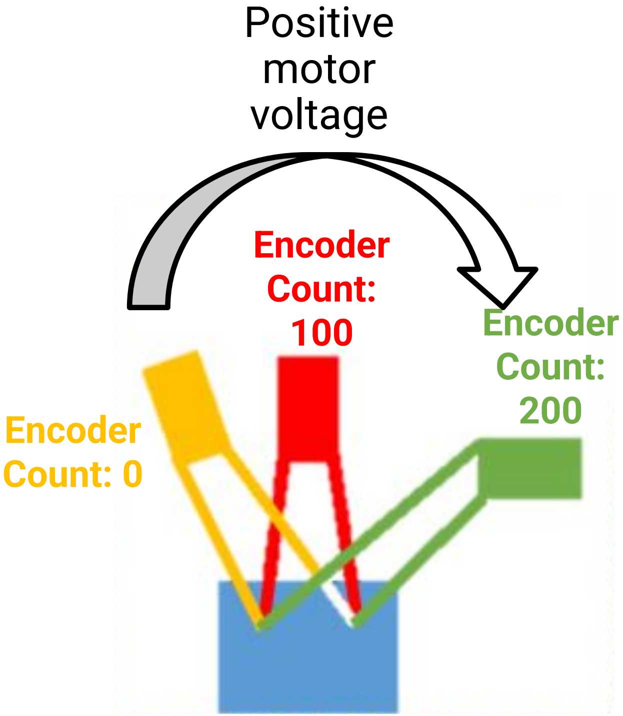
Friction Compensation

FRICTION_COMP_VOLTAGE: Arduino variable representing the motor voltage (V) required to overcome friction.

Increase **FRICTION_COMP_VOLTAGE** until linkage just barely moves.

FRICTION_COMP_VOLTAGE should be **positive** in your final code.

Example (note these are fake encoder counts):



Friction Compensation

```
// Compute the position error [encoder counts]
positionError = desiredPosition - motorPosition;

//** Feedforward terms: **//
// Compensate for friction.  That is, if we now the direction of
// desired motion, add a base command that helps with moving in this
// direction:
if (positionError < -5) {
  desiredVoltage = desiredVoltage - FRICTION_COMP_VOLTAGE;
}
if (positionError > +5) {
  desiredVoltage = desiredVoltage + FRICTION_COMP_VOLTAGE;
}
```

Note: This is in the final code (not the code used in lab today)

Mechanism Movement - Sign Convention

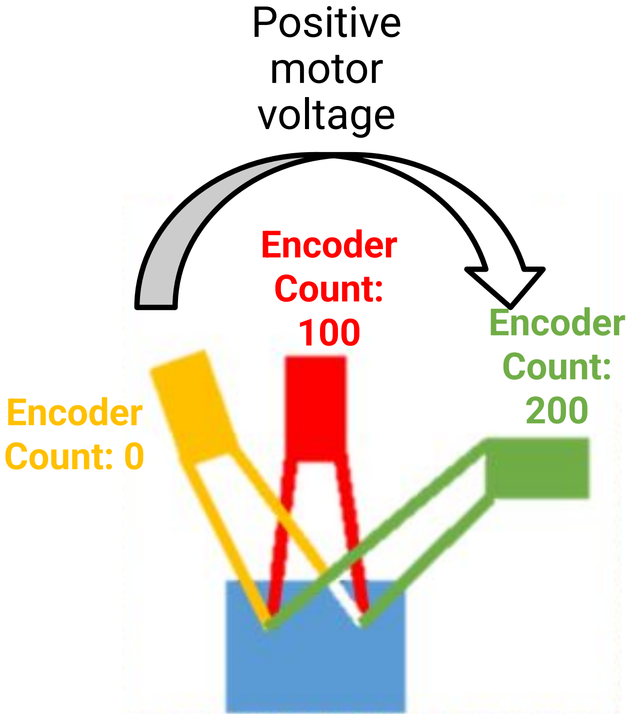
Encoder provides a measurement of the **position** of your linkage.

For controller to work:

- **Increasing encoder counts** should mean your mechanism is moving toward the right side of the playing field
- **Applying a positive motor voltage** (positive desiredVoltage Arduino variable) should make your mechanism move toward the right side of the playing field

__NOTE__: If this is not the case for either bullet point, you need to change your wiring (See Microcontrollers Tutorial #2)

Example (note these are fake encoder counts):



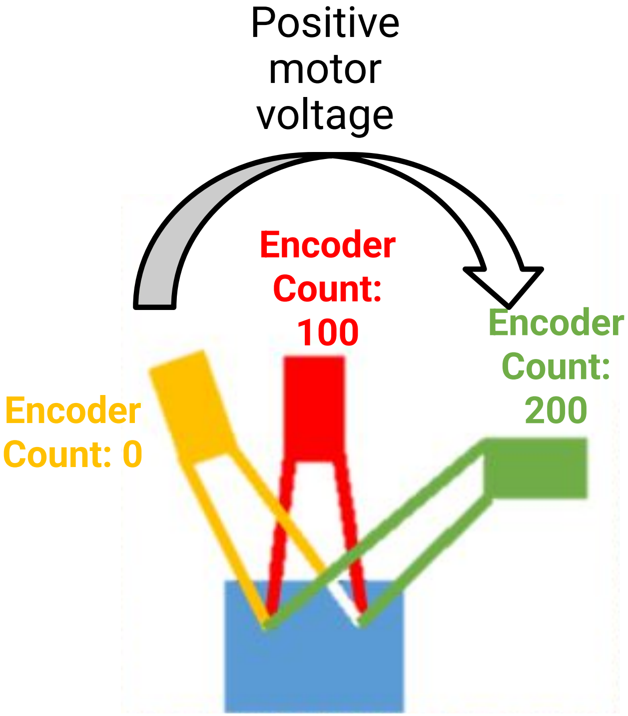
Target Position Setpoints

- **X_POSITION**: Arduino variable representing the *number of encoder counts* for each of the target positions
- Move the mechanism by hand and read the encoder output in the serial monitor
- Your values are unique to your mechanism and transmission

```
// CONSTANTS:
// Target positions:
const int TARGET_1_POSITION    = 0;
const int TARGET_2_POSITION    = 100;
const int TARGET_3_POSITION    = 200;
const int TARGET_4_POSITION    = 400;
```

CHANGE THESE VALUES

Example (note these are fake encoder counts):



Arduino Programming Pro-Tip

File > Preferences > “Use external editor”

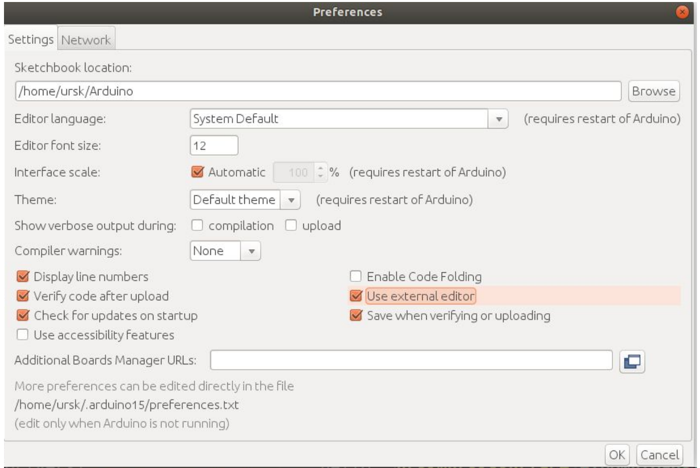
Then you can use another code editor!

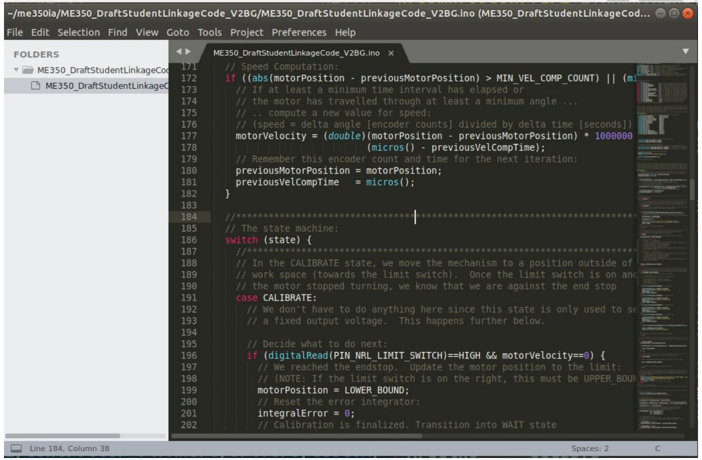
Our recommendations:

- Sublime Text
- VSCode
- Notepad++

Take advantage of better user interface such as indentation markers and tab auto-complete.

You still have to use Arduino to upload the code to the device.





Today

- Run through the Lab 9 Tutorial
  - Friction compensation
  - Test and set encoder counts
  - Complete Lab Assignment Worksheet 9

Due Next Lab

1. Complete Lab Assignment Worksheet 9
2. Add to wiring diagram